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EXPERIMENTAL AND NUMERICAL STUDY ON NOISE OF ELECTRIC VEHICLE CHARGING PILE

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ABSTRACT−This study investigates the sound source characteristics and propagation mechanisms of a 120 kW high-power direct current (DC) charging pile for electric vehicle (EV) charging pile through numerical simulations and experimental validation. Utilizing computational fluid dynamics (CFD) coupled with computational aeroacoustics (CAA), the acoustic spectrum and propagation behavior of the power module were analyzed. Experimental measurements confirmed simulation accuracy. Results indicate that the primary noise sources stem from dipole sources on internal cooling fan blades and adjacent component walls, with characteristic frequencies aligning with blade passing frequency (BPF) harmonics, demonstrating tonal dominance. The charging pile's acoustic spectrum mirrors the module's, with sound primarily radiating through vents in a directional pattern. A multi-factor noise reduction strategy was developed to address inherent propagation defects, achieving a 6.8 dBA reduction in overall sound pressure level (OASPL) at critical monitoring points. Acoustic experiments validated the optimized scheme's efficacy, highlighting its potential for high-power charging system noise control.

KEY WORDS : Electric vehicle, Noise optimization, Experimental validation, Numerical analysis, Charging pile

NOMENCLATURE

Hz : hertz (unit of frequency)

dBA: measurement unit used to express the ratio of one value of a physical property to another on a logarithmic scale

$\boldsymbol{Q}$: fluid flow volume

$\boldsymbol{P_s}$: the resistance of air flow through the system

SPL: sound pressure level

BPF: blade passing frequency

CFD: computational fluid dynamics

CAA: computational aeroacoustics

DES: detached eddy simulation

K-Omega SST: A turbulence model considering the transport effect of turbulent shear stress

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1. INTRODUCTION

Given the rising popularity of electric vehicles, the infrastructure supporting DC charging piles is rapidly expanding. DC charging piles offer high-power and fast-charging capabilities, rendering them essential for public charging needs. On one hand, the increasing demand for faster charging necessitates higher power levels, prompting the need for solutions to tackle the associated heat dissipation challenges. Conversely, improving heat dissipation often requires incorporating additional fans to enhance heat exchange, which in turn exacerbates aerodynamic noise problems. However, charging pile noise has become a major concern, as prolonged exposure to levels above 50 dB can cause health issues like stress, sleep disturbances, and reduced cognitive performance. In urban areas, where charging stations are often near homes, this noise pollution can significantly lower residents' quality of life. Additionally, the noise may deter potential users in quiet zones, such as residential or office areas, hindering electric vehicle adoption(Europe, 2011; Kogan *et al.*, 2018). As a result, mitigating the noise generated by charging piles has become a prominent area of research.

The power module, serving as the core functional component for the charging performance of a DC charging pile, constitutes the primary noise source, predominantly due to its internal cooling fan. Crucial for ensuring stable operation, power modules necessitate sufficient cooling, typically achieved through the installation of internal cooling fans. Nonetheless, the high-speed rotation of these fans generates aerodynamic noise primarily due to blade-air friction, vortex effects, and pressure differentials(Jang *et al.*, 2003). Scholars have studied the sound generation mechanism of fan aerodynamic noise, Sharland first showed that the noise of axial fan has dipole characteristics, and obtained its noise spectrum consists of broadband frequency and discrete frequency, which correspond to the turbulence noise and fan rotation noise(1964). Subsequent research in fan noise sources has focused on static and dynamic interference noise as well as vortex shedding noise(Longhouse, 1977), and Blake(Blake, 1986) summarized and classified the blade noise into interference noise and self-noise, thus providing a comprehensive understanding of fan noise generation. The advancement of computer technology has facilitated the adoption of computational aerodynamic acoustics as a key method for studying fan aerodynamic noise. The predominant prediction approach involves separately solving the flow and acoustic fields, filtering and decomposing fluid pressure from the flow field calculations to derive acoustic pressure, and subsequently employing acoustic models for field calculations(Lim *et al.*, 2016; Zanon *et al.*, 2018; Brentner and Farassat, 2003). Fans are commonly integrated as cooling components across various systems, prompting numerous scholars to employ numerical analysis and experiments to investigate aerodynamic noise from engineering fans in diverse settings. Cai-Ling Jiang used CFD and PIV experiments to analyze the flow field of an outdoor air-conditioning machine in detail, to obtain the source of aerodynamic noise of the complete machine, and calculated the aerodynamic noise index of the complete machine (Jiang *et al.*, 2006). Kai Ren simulated and measured that the main area of engine cooling fan noise is located at the tip of the blade edge(Ren *et al.*, 2023), and Jang-oh Mo numerically analyzed the engine cooling fan and also reached the above conclusions, but also concluded that the tip of the blade edge mainly generates tonal noise(Mo and Choi, 2020). These studies underscore how a combination of numerical analysis and experiments can effectively elucidate the aerodynamic noise characteristics of engine systems, aiding in overall noise reduction. Many researchers have explored fan blade optimization to diminish fan sound energy, Rama Krishna optimized the noise of motor cooling fans by changing the blade geometry, which effectively reduced the noise of the motor(Rama Krishna *et al.*, 2011), Gert Herold compared the noise test of different tilted forms of cooling fan blades and found that the noise of forward-swept blade is lower than that of backward-swept and non-swept fans(Herold *et al.*, 2017). Bionic blade technology for noise reduction has gained traction recently (Ma *et al.*, 2021; Wang *et al.*, 2021; Sundeep *et al.*, 2023). Nevertheless, optimizing entire air ducts becomes imperative in systems where individual fan blade optimization is unfeasible due to the fan always purchased for the whole system. Jie Tian reduced the broadband noise of an air conditioner by adjusting the tip angle of the grille(Tian *et al.*, 2009), Junwei Hu compared the effect of deflector ring width on the aerodynamic noise of an air conditioner to obtain the optimum deflector ring width to reduce the turbulence noise(Hu and Ding, 2006), and Sun and Takushima proposed to optimize the upstream and downstream flow patterns of fans by installing deflectors upstream and downstream of the fans to reduce the airflow noise(Sun *et al.*, 2021; Takushima *et al.*, 1992). While Wang observed that installing a rectifier upstream of a fan can reduce broadband noise, but it is not obvious to reduce the rotational noise of the blades(Wang, 2018). The broadband noise caused by turbulence in the whole system accounts for a smaller percentage compared to the blade rotational noise, so to effectively diminish overall system noise, efforts must focus on attenuating blade rotational noise propagation as a primary strategy over turbulence-induced broadband noise reduction.

Given the complexity of the internal system of the charging pile and the multitude of fans, predicting and controlling the noise of the entire system presents numerous challenges. Existing studies on charging pile noise control have predominantly focused on isolated measures–such as power limitation, sound barriers, or inlet/outlet silencing elbows(Clar-Garcia *et al.*, 2025; Bang *et al.*, 2019)–yet lack a holistic approach to balance thermal performance and acoustic optimization. This study integrates the physical structure of the charging pile with a thorough analysis of its acoustic characteristics using numerical and experimental methods. It aims to optimize air ducts to achieve a balance between heat dissipation and attenuation of sound sources. Initially, a computational fluid dynamics (CFD) coupled with computational aeroacoustics (CAA) calculation method was employed to analyze the noise spectrum and sound field distribution characteristics of the power module, which serves as the primary sound source within the pile. Concurrently, experimental measurements were conducted to validate the accuracy of simulations. Subsequently, the sound propagation of the charging pile was analyzed, revealing its propagation characteristics and spectral properties. Control of the sound propagation from the fan source is achieved through the design of optimized air ducts.

Unlike previous work that prioritizes either thermal management or noise suppression, our approach synergistically addresses both challenges by redesigning airflow paths to minimize radiation noise while maintaining cooling efficiency.Subsequently, acoustic tests were conducted to verify the effectiveness of the optimized scheme in reducing the radiated noise from the charging pile.

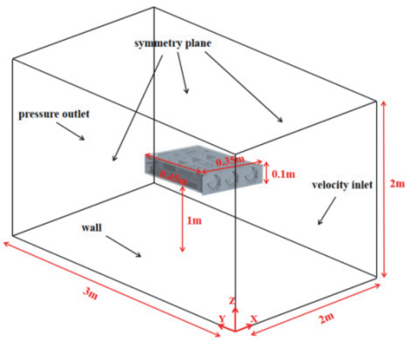
2. ACOUSTIC CHARACTERISATION OF SOUND SOURCE COMPONENTS FOR CHARGING PILES

2.1. Flow field analysis of power module

This paper initiates by analyzing the sound source characteristics and the laws governing sound propagation within a single module, establishing a fundamental groundwork for the comprehensive acoustic analysis of the charging pile. Figure1(a) illustrates the computational domain and dimensions of the comprehensive flow field model of a single power module, which replicates the full-load operation of the power module within a semi-anechoic chamber, operating at a fan speed of 11,000 revolutions per minute(RPM). To achieve a more precise analysis of noise generation and propagation in the fan installation system, the structural models of all internal devices of the power module were retained, albeit with some degree of simplification to enhance regularity.

The core region of the module was meshed using a polyhedral approach, featuring five prismatic layers aligned perpendicular to the fan and the module wall’s normal direction. These layers specifically addressed flow characteristics within the surface layer, effectively encapsulating the rotating domain and the narrow regions inside the internal air duct. In the far-field region, the trimmer mesh configuration was adopted, expanding the mesh size incrementally to streamline solution time without compromising accuracy. The fluid mesh layout of the module is depicted in Figure1(b).

(a)



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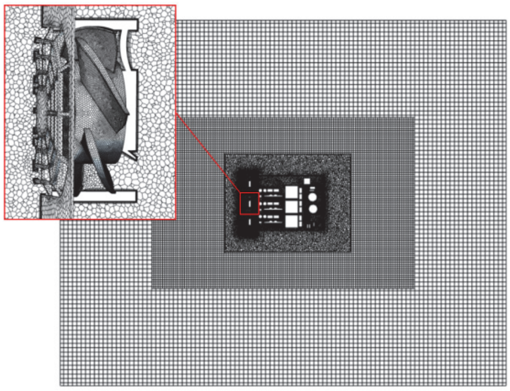


Figure 1. CFD model of the power module: (a) Fluid domain; (b) Fluid mesh.

The surface of the computational domain facing the inlet of the module was designated as the velocity boundary, the surface facing the outlet of the module was designated as the pressure boundary, the surface one meter away from the bottom of the module was designated as the wall boundary, and the rest of the surfaces were designated as the symmetric plane boundary. A moving reference system was employed to simulate the rotation of the fan area, and an interface was utilized for data transfer between different areas. The K-Omega SST turbulence model was initially employed to solve the steady-state flow field of the model, with the results of the steady-state flow field served as the initial conditions for the transient flow field solution. This approach facilitates the convergence of the transient flow field solution. The transient turbulence model adopted the DES model, incorporating module sound source data as the excitation term for subsequent sound propagation calculations. Adhering to Nyquist’s sampling law, a transient sampling time step of 1.7×10^{-4}s was established, resulting in a total sampling time step of 0.2s. Maintaining flow field stability required that the transient solving unit time step be lower than the sampling time step. Considering the fan rotation cycle every 3° as a time step equivalent to 4.5×10^{-5}s, and the total time step was set at 0.24s, allowing for several cycles of fan stabilization before sampling to be conducted for a more stable flow field.

(a)

(b)

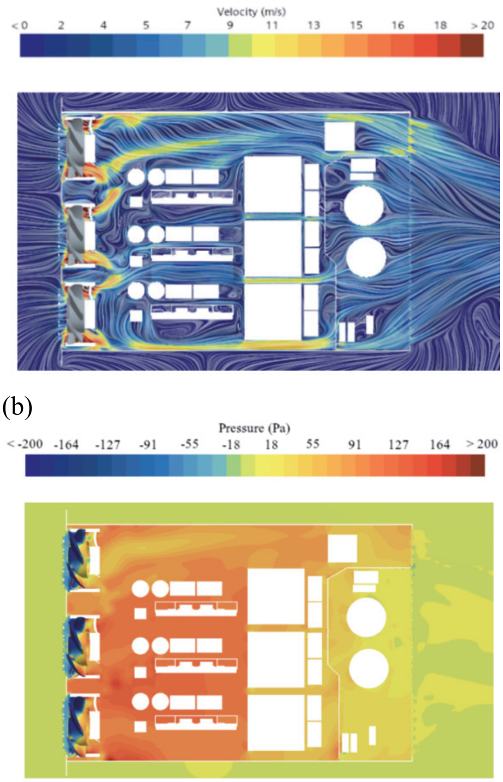


Figure 2. Flow field map of the power module: (a) Velocity map; (b) Pressure map.

Firstly, the flow field map of the power supply module is presented, with the velocity field distribution shown in Figure 2(a). It can be observed that the high-speed fluid is concentrated in the local area of the fan's air outlet surface, with the maximum wind speed in this area reaching 20 m/s. The device's relatively high density results in a substantial wind resistance, impeding the downward flow of high-speed gas from the fan outflow. This gas is instead redirected to the device gap area, where it forms a large vortex. This vortex also serves to generate turbulence noise. The continuous operation of the fan creates a pressure differential between the upstream and downstream sections, as illustrated in Figure 2(b). The high-velocity wind exerts a compressive force on the airflow from the fan outlet to the downstream section. Upon striking the device positioned in the direction of the fan outlet, the compressed high-speed gas generates a high-pressure fluctuation on the surface, which is also the underlying cause of the turbulence noise. The airflow pressure diminishes in direct proportion to the distance downstream, due to the resistance of the device. The airflow out of the module outlet is essentially at the same pressure as the outside air pressure.

2.2. Aerodynamic acoustic source analysis of power modules

The aerodynamic noise of the fan represents the primary source of noise in the power module. The high-speed rotation of the blades is the primary factor affecting the aerodynamic noise of the fan. The fan speed and the sound pressure level (SPL) exhibit a specific linear relationship, as indicated by the following equation:

$$SPL_{N_2} = K_1 * log(\frac{N_2}{N_1}) + SPL_{N_1} \tag{1}$$

Where K_1 is the empirical coefficient, generally small axial fans take the value of 55, SPL_{N_2} is the sound pressure level at the target fan speed,while SPL_{N_1} represents the SPL at the fan reference speed.The measured point SPL of the fan at different rotational speeds were calculated using the above equations. Further numerical simulations were carried out using the FW-H integration method(Cheng *et al.*, 2024; Ji and Liu, 2012) to predict the variation of the SPL at the measurement point, with experimental data used for verification, as illustrated in Figure 3.

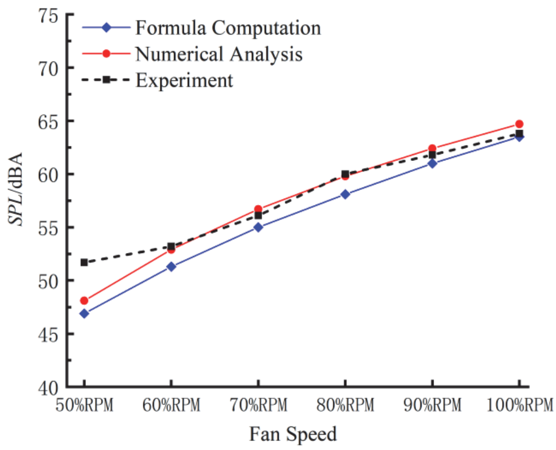


Figure 3. Comparison of formula computation, numerical analysis and experiment of SPL with fan speed.

Figure 3 illustrates that the fan noise increases significantly with rising rotational speed. This phenomenon is attributed to the fact that increased speed leads to greater pressure fluctuations in the fan blades, which in turn enhances the radiation of sound pressure. The experimental data and simulation results demonstrate that the increase in SPL is linear when the fan speed is increased from 50% to 100%RPM. However, the increase in SPL is more pronounced in the mid to high speed range, particularly between 60% and 80% RPM, while the rate of increase decelerates in the range between 80% and 100% RPM. The simulation result demonstrates a high degree of correlation with the experimental data in predicting SPL at all RPMs. In contrast, the empirical formulae exhibit a general tendency to produce lower values than the simulation and experimental results, suggesting that they may have limited efficacy in accurately predicting SPL.

In order to satisfy the thermal management requirements, the fan frequently requires continuous operation at the maximum design speed to adapt to the airflow requirements of the various systems. Given the significant impact of system resistance on fan ventilation volume and noise level, the relationship curve of the law of the fan ventilation volume Q and SPL with the system resistance P_s in Figure 4 has been plotted using simulation calculations. As illustrated in Figure 4, the simulation and experimental data demonstrate that the fan ventilation volume Q declines as the system resistance P_s increases. This phenomenon can be attributed to the fact that the fan, when confronted with a higher resistance, must invest more energy to overcome that resistance, which subsequently results in a reduction in efficiency and a decrease in the energy allocated to air movement, thus leading to a reduction in the airflow.

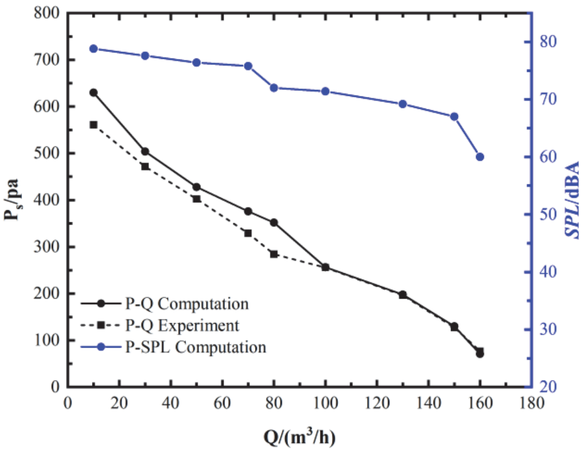
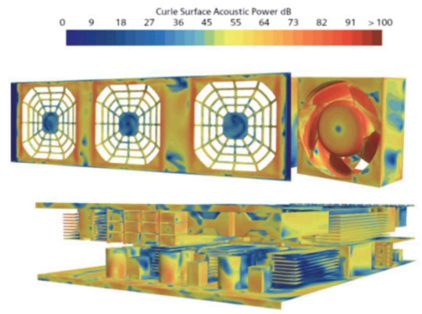


Figure 4. The law of the fan ventilation volume Q and SPL with the system resistance P_s .

Further analysing Figure 4, the SPL of the fan increases with the increase of the system resistance. This trend can be explained by the fact that a larger system resistance leads to a higher pressure on the fan blades, which exacerbates the pressure pulsation on the blade surface and thus enhances the sound pressure radiated . Therefore, in the system design, the system resistance should be reduced as much as possible to keep the fan noise at a low level.

In power module systems, the primary sources of aerodynamic noise are periodic pressure fluctuations caused by fan blade rotation and the generation, break-up and separation of eddies in downstream turbulence. The former is typically regarded as a dipole surface acoustic source, while the latter is considered a quadrupole volume acoustic source. The following figure illustrates the distribution of the two sound source intensities in the power module system. Figure 5(a) illustrates the distribution of the surface dipole sound source across the surface of each component in the module system. In particular, the surface dipole sound source intensity is higher in the upstream and downstream areas of the fan, which experiences high flow velocity. The sound source on the fan blade is the largest, due to the surface pressure pulsation sound source generated by the fan blades periodically beating the surrounding air. In some lower flow areas downstream, the air flow is insufficient to effectively provoke this surface pressure pulsation, resulting in a low intensity. Figure 5(b) illustrates that the intensity of the quadrupole sound source within the module system is considerably less than that of the dipole sound source. This is due to the relatively low velocity of the airflow generated by the fan and the downstream duct is complex and narrow, which impedes the development of turbulence to a greater extent. Consequently, the quadrupole sound source in the downstream duct of the module as a whole is low, and it is mainly concentrated in a localized area close to the fan air outlet.

(a)



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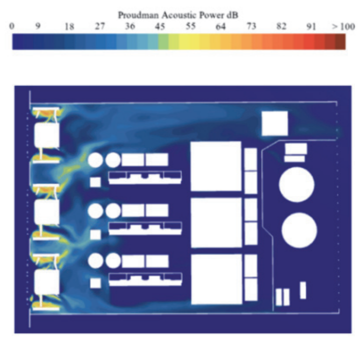


Figure 5. Acoustic source maps of the power module: (a) Acoustic dipole source; (b) Acoustic quadrupole source.

2.3. Acoustic propagation analysis and verification of power modules

In order to further analyze the aerodynamic noise propagation characteristics of the module, The acoustic computational model for the module was established, as illustrated in Figure 6(a), and the module semi-infinite element acoustic model must be established in accordance with the semi-anechoic chamber. Acoustic finite elements were then employed to simulate the near-field sound propagation. This encompasses the reflection and refraction of sound waves, as well as the superposition and cancellation of sound waves. In order to predict the far-field sound propagation, an acoustic infinite element surface was employed. It is assumed that sound waves propagate uniformly outward in the region of the infinite element surface and beyond. A fully reflective rigid wall boundary was employed to simulate ground reflection effects. The hemispherical envelope method of the ISO3744 international standard was utilised for the calculation of the modular sound power, with experimental measurement conducted to verify the accuracy of the simulation.

(a)

(b)

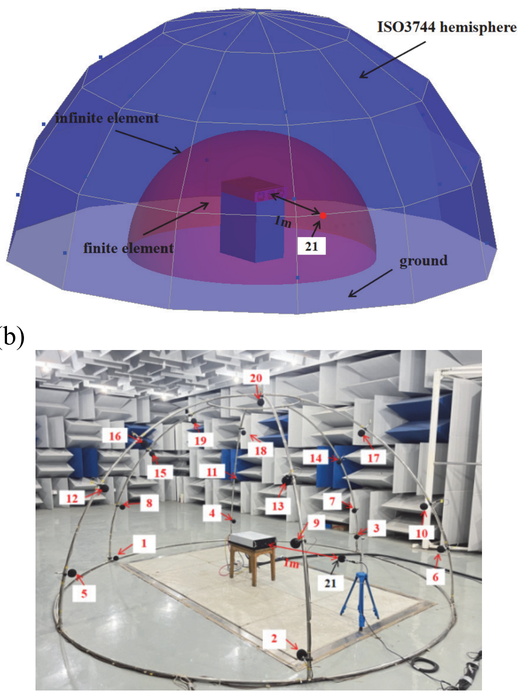
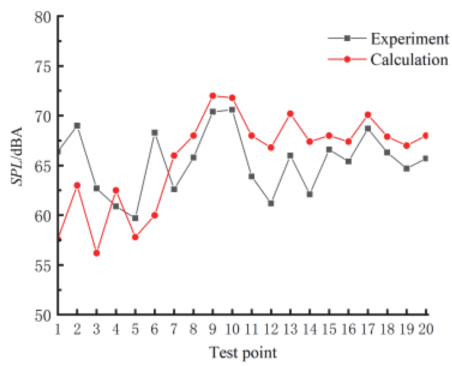


Figure 6. Power module computational model and experimental mode: (a) Computational model; (b) Experimental site and microphone layout.

The sound source data obtained from the aforementioned transient flow field solution was employed as the excitation for the direct frequency response calculation, with the sound source excitation encompassing both the dipole surface source and the quadrupole volume source. This was achieved by using the CFD coupled CAA calculation method(Huang *et al.*, 2021).The acoustic simulation of the power module, along with the corresponding measured results, are presented in Figure 7 below. Initially, the OASPL at each monitoring point is derived from the test results, as illustrated in Figure 7(a). It can be observed that monitoring points 9 and 10 have the highest OASPL, which were located in the area directly in front of the air inlet of the module. Similarly, the OASPL at monitoring points 13 and 17 are also relatively prominent, and they were located in the area directly above the air inlet of the module. This indicates that the noise from the power supply module mainly radiates in the direction of the air inlet in the shape of a trumpet, with a strong directivity. The measured results are in good agreement with the simulation results. The measured OASPL are also of note at monitoring points 2 and 6, which were situated directly below the module air inlet. Due to the proximity of this region to the ground, the interaction of sound waves in this region is complex, resulting in numerous wave peaks and troughs. When the simulation monitoring point was located precisely at the trough position, the OASPL of the monitoring point is underestimated, thereby introducing a bias in the simulation results compared to the measured values.

(a)



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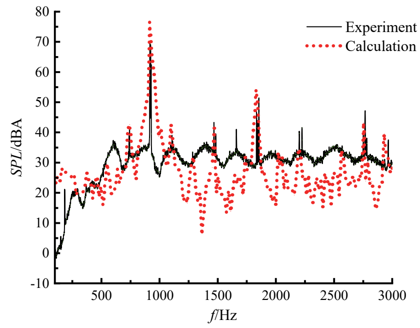


(c)

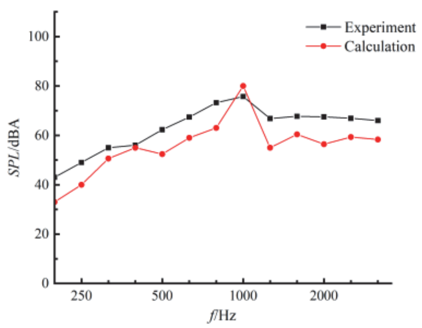


Figure 7. Comparison of power module simulation and measurement results: (a) Comparison of OASPL at each monitoring point; (b) Narrow-band spectrum comparison of SPL at monitoring point 21; (c) The 1/3 octave comparison of sound power levels.

Monitoring point 21, situated one metre from the module inlet, was selected for sound pressure level(SPL) spectrum analysis, as illustrated in Figure7 (b). The simulation results demonstrate that the peak characteristics of the main frequency in the spectrum are particularly pronounced. The main frequency is the BPF at 917 Hz, and other peak frequencies appear at the harmonics of the BPF. These include the first-order and second-order harmonics at 1834 Hz and 2751 Hz. These frequency peaks are discrete tonal noise caused by the fan blades periodically beating the surrounding air. Concurrently, the formation of other frequency amplitudes within the narrow band spectrum is attributed to the influence of the pressure pulsating sound source on the component wall and the downstream turbulent sound source. In conclusion, the aforementioned sound sources collectively define the aerodynamic noise spectrum characteristics of the modular system. The spectral characteristic of the measured points is entirely consistent with the simulation results, and the amplitude of the characteristic frequency is also consistent with the simulation results, which further verifies the accuracy of the simulation.

The calculation of the sound power level according to the hemispherical envelope surface method in ISO 3744 yielded the one-third octave spectrum of the module sound power, as illustrated in Figure 7 (c).The frequency band with a center frequency of 1000Hz is the main sound energy band. The simulation results of this frequency band are very close to the measured ones, as this frequency band contains the BPF and the main energy of this frequency band is also contributed by the amplitude of the BPF. The simulation and measurement results demonstrate that the SPL amplitude of the BPF at the monitoring point is in good agreement, indicating that the error in the simulation and measurement amplitude of the sound power level in this frequency band is minimal. However, in different frequency bands, the majority of the sound energy of each frequency is attributed to turbulence noise, and the simulation has inherent limitations in predicting the actual eddy current, resulting in discrepancies between the simulation and measurement in these frequency bands.

(a) (b) (c)

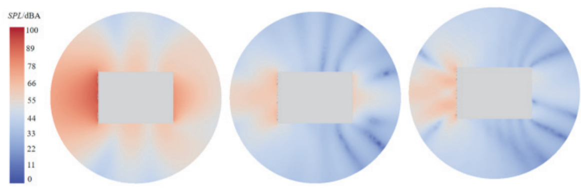


Figure 8. Acoustic pressure distribution maps of the power module: (a) SPL distribution maps of the power module at 917Hz; (b) SPL distribution maps of the power module at 1834Hz; (c) SPL distribution maps of the power module at 2751Hz.

The axial section of the fan within the power module is depicted in the accompanying figure, illustrating the distribution of SPL. A detailed analysis has been conducted on the SPL, focusing on the BPF as well as its first and second harmonic frequencies. As delineated in Figure 8, the sound waves are observed to emanate with marked directivity through both the inlet and outlet of the module. Notably, the region proximal to the module's inlet is subjected to the most intense sound pressure radiation. Additionally, the sound pressure is observed to radiate with varying intensities across the upper and lower 90-degree sectors relative to the axis of the inlet, manifesting a distribution pattern reminiscent of a trumpet. At different frequencies, the radiation energy of sound pressure to the inlet wind side is greater than that of the outlet wind side. At the non-inlet and outlet wind side of the module, the SPL is relatively low due to the acoustic diffraction effects. When sound waves arrive at these areas, they lose more energy and experience more mutual interference, which results in the formation of the trough area observed in the figure. In particular, at the higher frequency stage, the trough area is further enhanced. The acoustic simulation and experiment of the single module allows for the understanding of the sound source characteristics and propagation rules of the module system. This provides the basis for the subsequent study of the sound propagation characteristics of the charging pile.

3. ACOUSTIC PROPAGATION ANALYSIS OF CHARGING PILE

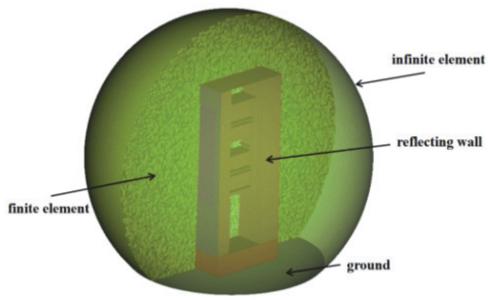
3.1. Numerical simulation of charging pile acoustic propagation

In this study, a 120kW DC charging pile was selected as the subject of investigation. The charging pile is equipped with four 30kW power modules, resulting in a total power output of 120kW. Typically, the full-power operation condition is the most representative working state of charging piles, especially in fast-charging scenarios where charging piles usually operate at maximum power to meet users' demand for rapid charging. Therefore, the noise characteristics under full-power operation can reflect the acoustic performance of charging piles under the most representative working conditions. Hence, this paper adopts the full-power operation condition of charging piles as the noise evaluation criterion and conducts noise reduction optimization based on this condition, providing the most stringent design basis for noise control.

Due to the numerous fans present in the power modules of the charging pile, it is challenging to compute the sound propagation issue using the conventional CFD coupled CAA computational method. Considering the relatively small size of the sound source in the modules, an acoustic evaluation in the far field of sound propagation is usually required for the entire charging pile system. At distances significantly greater than the source size, the sound source can be approximated as a point source. Therefore, the sound source of a single module was treated as a point source, and the spectrum of the source obtained from the measurement was utilized as an excitation parameter to conduct sound propagation calculations for the entire charging pile system.

The acoustic finite element model for the entire pile was established, and the acoustic grid for the pile is depicted in Figure 9(a). According to the theory of plane wave acoustic grid scale, each wavelength should have a minimum of 6 grid nodes to ensure the accuracy of numerical calculations for the acoustic field. In this study, the acoustic frequency was calculated up to 3000Hz, indicating that the grid scale should be less than 20mm.To avoid the influence of ground reflection on the SPL at the monitoring points and to ensure proximity to the human ear, sound pressure monitoring points (p_1, p_2, p_3, and p_4) were set up at heights of 1m and 1.5m from the pile body. Among these points, p_1 and p_2 were placed on the inlet and outlet wind sides of the pile, as shown in Figure 9(b). These points were selected to evaluate the distribution of the SPL for the entire pile and to facilitate subsequent comparisons with measurements. At a distance of ten meters from the pile, additional acoustic monitoring points were positioned at 10° intervals around the circumference of the pile, as illustrated in Figure 9(c). These points were used to assess the acoustic far field directivity of the pile.

(a)



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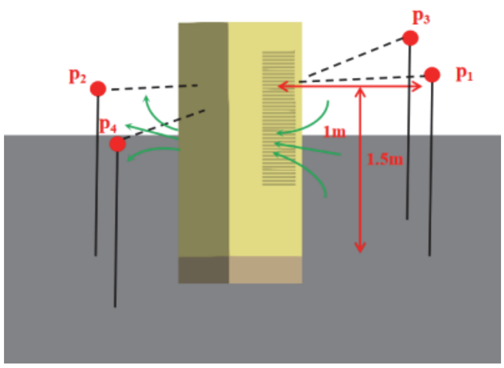

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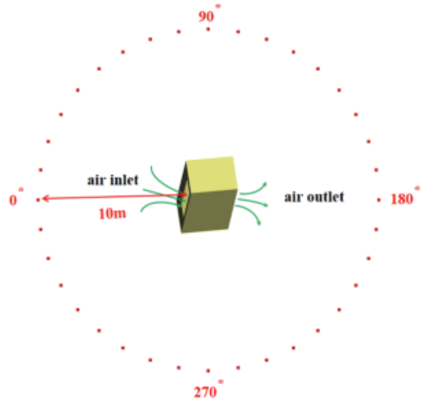


Figure 9. Schematic diagram of charging pile acoustic finite element model and monitoring point arrangement: (a) Acoustic finite element model; (b) Sound pressure monitoring point; (c) Directivity monitoring point.

In Figure 10(a), the simulation SPL spectrum is depicted for four distinct monitoring points across the charging pile under full-power conditions of the module. Across the entire frequency range, the spectral patterns at each monitoring point exhibit fundamental consistency. The dominant frequency observed corresponds to the BPF of the module' s fan, accompanied by its harmonic component. This harmonic frequency mirrors the spectrum observed for a single module, suggesting that the structural integrity of the pile body does not alter the inherent frequency characteristics of the module' s sound source. Notably, peak values of both the BPF and its harmonic are observed to be elevated compared to those of a single module, indicating that the sound sources from multiple modules within the pile exhibit additive amplitudes at their peak frequencies.

Figure 10(b) portrays the sound radiation directivity for the entire pile at BPF. This distribution highlights significant directivity in the far-field sound propagation from the pile. The module sound sources primarily radiate outward from the air inlet and outlet of the pile. The sound radiation pattern on the inlet side exhibits a larger and broader range, attributable to the closer proximity of the module sound sources to the air inlet. Conversely, the side of the pile not aligned with the air inlet and outlet effectively shields the propagation path of the modular sound source. This side is predominantly influenced by the diffraction of sound waves emanating from the air inlet and outlet, resulting in a comparatively lower total sound energy.

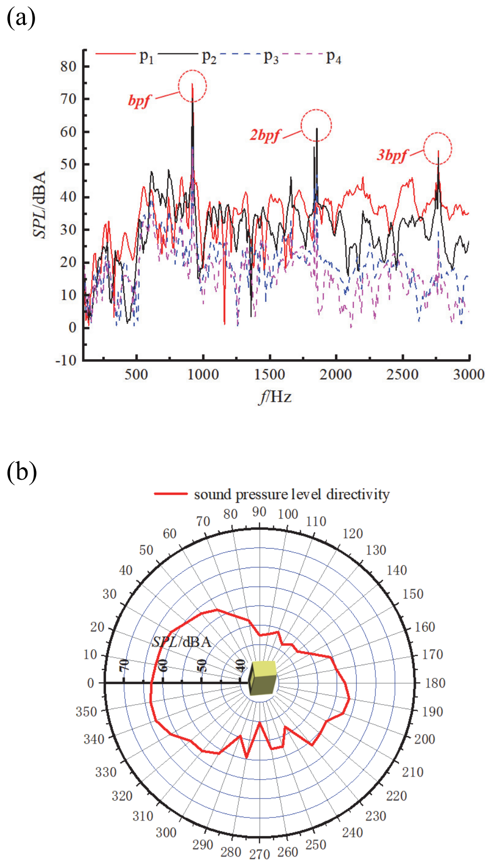


Figure 10. Acoustic simulation results at monitoring points: (a) Spectrum of SPL of simulation results at monitoring points; (b) Sound radiation directivity pattern of simulation results.

In Figure 11, it is evident that the module' s inlet and outlet directly link to the vent of the charging pile. This setup enables the module' s sound source to propagate outward directly through the charging pile' s inlet and outlet, with minimal obstructions in the primary propagation path. While such an air duct configuration facilitates module heat dissipation, it also poses a significant noise challenge. Typically, ventilation louvres are installed at the charging pile' s inlet and outlet, which can partially impede sound propagation at higher frequencies. As in Figure 11(c), when the sound wave frequency exceeds twice BPF, At this time the frequency is 2751Hz, the wavelength becomes smaller than the louvre size, resulting in increased reflection loss between the louvres. However, at lower frequencies, especially at the main frequency, the louvres are less effective at blocking the main frequency sound waves' propagation, as shown in Figure 11(a). Therefore, to manage noise propagation from the charging pile, it is essential to implement appropriate obstruction design on the highly directional sound source propagation path.

(a) (b) (c)

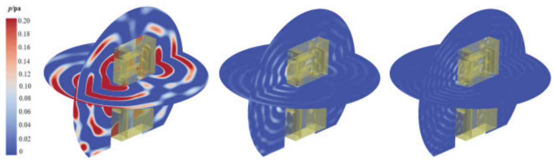
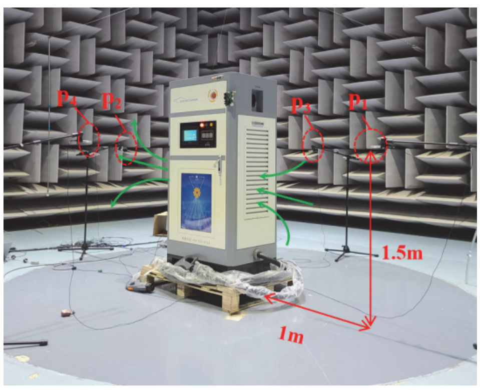


Figure 11. Sound pressure field distribution of the pile at characteristic frequencies: (a) Sound pressure field distribution of the pile at 917Hz; (b) Sound pressure field distribution of the pile at 1834Hz; (c) Sound pressure field distribution of the pile at 2751Hz.

3.2. Experimental validation of the charging pile acoustic propagation

The experiment was carried out in a semi-anechoic chamber, and each microphone placement is shown in Figure 12(a), and the location of the microphone array corresponds to that of the simulation monitoring point, so that the verification of the alignment is carried out, and is also used to evaluate the acoustic wave propagation characteristics of different piles in each direction.

(a)



(b)

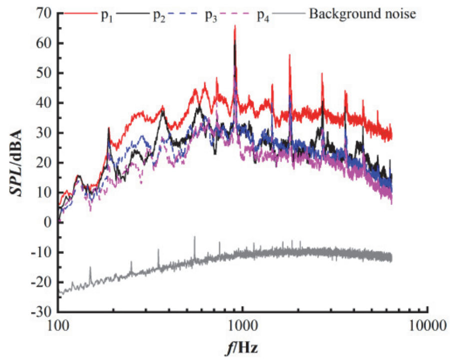


Figure 12. Experimental setup to measure noise and results: (a) Experimental site and microphone layout; (b) Experimental Spectrum of SPL at monitoring microphones.

Under the full power operating condition of the module, The SPL spectrum of the original model at each monitoring point obtained in Figure12(b), The spectral profiles at each monitoring point exhibit a high degree of consistency. Their primary frequency aligning with the BPF, while secondary eigenfrequencies are predominantly distributed among the harmonic frequencies of the BPF across the entire frequency spectrum. Monitoring point p_1 exhibits the highest SPL, followed by point p_2, whereas point p_3 and point p_4 consistently register lower overall levels. This observation suggests that the pile emits the highest sound pressure in the vicinity of the air inlet, with significantly lower levels on the non-vent sides, displaying distinct directional characteristics consistent with simulation predictions. Additionally, two specific monitoring points, p_1 and p_2, situated along the primary propagation axis of the pile noise, were selected for simulation validation, as they represent pivotal locations for assessing the acoustic performance of the pile.

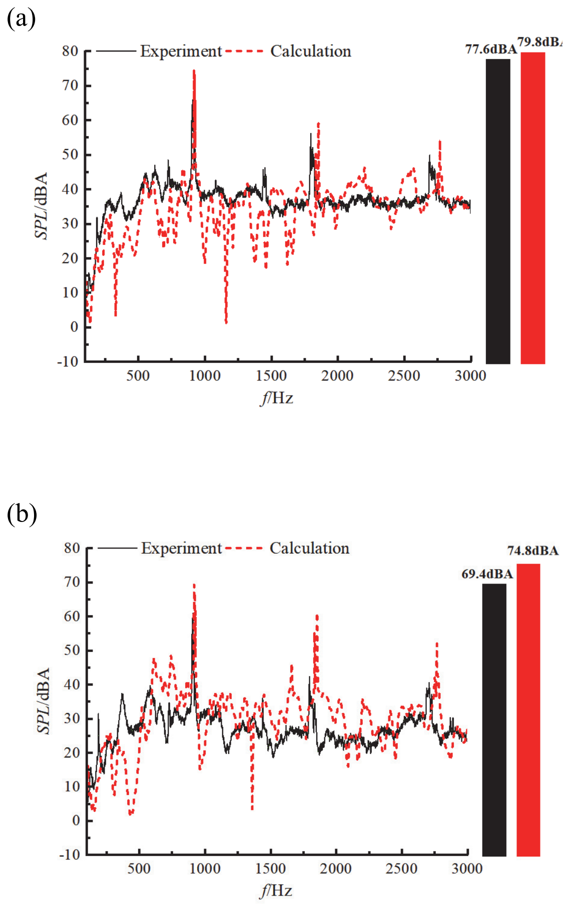


Figure 13. Comparison of SPL spectrum obtained by simulation and measurement at major points: (a) Comparison at point p_1; (b) Comparison at point p_2.

Figure13 shows the sound spectrum of the pile obtained by the numerical analysis and experimental measurement at p_1 and p_2 under the full power operating condition of the module. It can be observed that the characteristic frequency distribution law of the simulation is consistent with the measured results, but there is a certain delay in the BPF and its harmonic frequencies. The main reason is that the number of modules in the whole pile is large, and it is difficult to control the fan speed of all the modules accurately. For instance, during testing, the blade frequency was recorded at 910Hz, which corresponds to an actual fan speed of approximately 10920 RPM. In contrast, the simulation input was based on the BPF of 917Hz with a full rotation of 11000RPM, resulting in a noticeable hysteresis effect. This effect is particularly pronounced in the harmonic frequencies, especially at the double and subsequent multiples. In addition, for the main frequency peaks, we observe that the simulated values are always higher than the measured values, because the simulation is better at superimposing the peaks of the BPF and its harmonic frequencies for multiple sources, and therefore overestimates the peaks of the BPF and its harmonic frequencies. In terms of overall sound pressure level (OASPL), the simulated value is 79.8 dBA and the measured value is 77.6 dBA, with an error of 2.2 dBA at point p1; the simulated value is 74.8 dBA and the measured value is 69.4 dBA at point p_2, with an error of 5.4 dBA. The larger error in OASPL at measurement point P_2 stems from insufficient consideration of directional characteristics when modeling the module as an equivalent point-source. As demonstrated in Section 2.3, the module exhibits dipole-like radiation patterns with higher sound pressure in the intake region than the exhaust region. The point-source equivalence overestimates acoustic source strength in other directions, leading to a larger error in the OASPL at p_2.

By comparing the acoustic simulation of the pile with the experimental results, it can be concluded that the simulation method which equates the sound source to the point sound source can effectively predict the sound propagation law of the pile, and the sound pressure prediction at the sound propagation monitoring point is also more accurate.

4. ACOUSTIC OPTIMIZATION AND VERIFICATION OF THE CHARGING PILE

4.1. Design and analysis of the pile acoustic optimization scheme

The necessity for waterproofing and dust proofing of charging piles gives rise to the integration of structures such as dustproof cottons and shutters, which in turn increases the resistance of the system. Therefore, optimising noise reduction at the system resistance is a challenging endeavour. In contrast, achieving the pile noise reduction by blocking the noise transmission path is a more appropriate approach. However, the quality of the noise reduction barrier design will directly affect the thermal performance of the power module. Therefore, it is necessary to balance the heat dissipation and noise reduction in the design process to ensure that effective noise reduction optimisation is carried out while meeting the heat dissipation requirements.

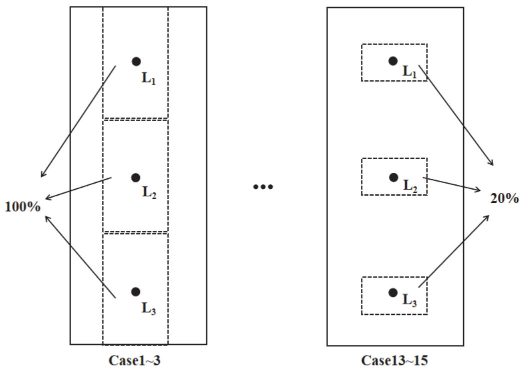


Figure14. Schematic diagram of the optimised design scheme.

This study considers the ventilation area A and the ventilation location *L* of the pile as the key influencing factors, with a detailed optimisation analysis being carried out. In consideration of the ventilation area *A*, five levels are established. These levels are based on a gradual reduction of the original pile ventilation area *A* by 20%. This reduction corresponds to the following levels: 100%, 80%, 60%, 40%, and 20% of the original area. In order to accommodate the limited width of the door panel, the width of the vent is maintained, while the height is altered to adjust the size of the ventilation area. Three variable positions of *L* have been established,these are designated L_1, L_2 and L_3, and correspond to the upper, middle and lower positions of the inlet and outlet of the pile. These positions is determined based on the centroid of the original ventilation area at the upper, middle and lower vents. Ensuring that the centre point of the vent at each location is fixed, when the ventilation area is altered, and that the relative positions of L_1, L_2 and L_3 remain fixed. This design is intended to ensure consistency and comparability of ventilation locations. The optimal design scheme is shown in schematic form in Figure 14, and Table 1 shows the optimal design full-factor, full-level scheme to develop a total of 15 cases.

Table 1. Summary of design scheme.

Case	Factors		Results	
	A	*L*	ΔT/°C	*SPL*/dBA
1	100%	L_1	11.6	87.9
2	100%	L_2	11.5	84.2
3	100%	L_3	12.2	82
4	80%	L_1	12.1	81.9
5	80%	L_2	12	84.2
6	80%	L_3	12.9	76.6
7	60%	L_1	12.4	77.5
8	60%	L_2	12.1	81.9
9	60%	L_3	13.1	73.5
10	40%	L_1	13.7	75.9
11	40%	L_2	13.1	74.5
12	40%	L_3	14.1	70.4
13	20%	L_1	17.7	70.5
14	20%	L_2	16.9	71.4
15	20%	L_3	17.7	65

The numerical simulation analysis of the air temperature rise and noise level inside the pile is conducted for each scheme presented in Table 1. The simulation results are presented in Table 1 and Figure 15. The results demonstrate a clear trend of increasing air temperature and decreasing noise level within the pile as a consequence of the reduction in ventilation area. In particular, it was found that while maintaining the same ventilation area, the vents at the bottom of the pile achieved the lowest SPL and the highest temperature rise. This phenomenon can be attributed to the fact that the layout of the bottom vent effectively extends the propagation path of the sound source and completely blocks the main propagation direction of the sound source, thus significantly increasing the attenuation during the propagation of sound waves and effectively reducing the propagation of noise. However, this layout also causes the air flow to undergo multiple deflections as it flows through the interior of the pile, significantly increasing the flow resistance of the system and thereby affecting the heat dissipation performance.

In accordance with the technical specifications of the power module, the optimal operational temperature range is -40° C to 75° C. To ensure the long-term stable operation of the module, it is essential to maintain the internal temperature of the pile where it is situated within this specified range. Additionally, given that the pile may operate in an extreme ambient temperature of 50° C, it is essential to keep the temperature rise of the air within the pile to below 25° C. Since the design does not account for0 ventilation resistance from components such as filter cotton and louvers, the maximum temperature rise of the air within the pile should not exceed 15° C.

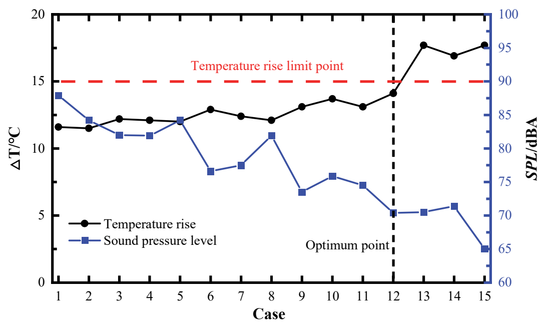
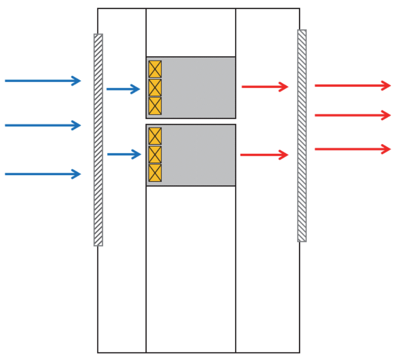


Figure 15. Temperature rise and SPL of all design scheme.

Consequently, upon evaluating the temperature rise and SPL data across all design schemes, it is ascertained that Scheme (12) is the most optimal solution. This scheme features a ventilation area that is reduced to 40% of that in the original scheme, with the ventilation duct positioned at the lower of the pile. Figure 16 depicts the schematic diagrams of the pile air duct for both the original model and the most optimal scheme (12). To align with practical application requirements, the final optimized Scheme (12) incorporates honeycomb holes are arranged in the ventilation system, thereby enhancing resistance to dust and water ingress while maintaining the ventilation area in accordance with the design specifications. Furthermore, to facilitate the unimpeded flow of air within the pile, a deflector has been introduced to refine the directionality of the airflow, as illustrated in Figure 16(b). In this figure, the blue arrows denote the influx of external cold air, while the red arrows represent the egress of hot air following heat transfer, with the direction of the arrows indicating the prevailing direction of airflow.

(a)



(b)

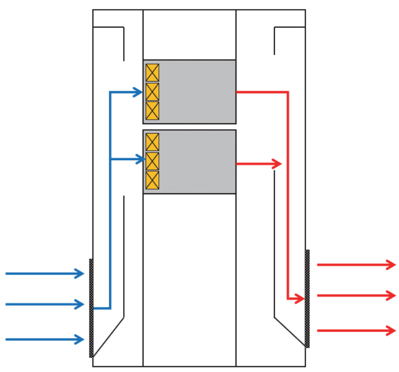
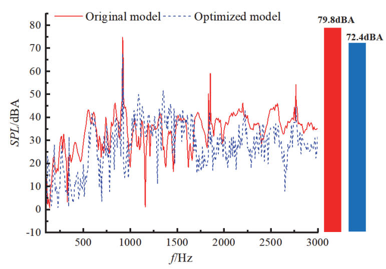


Figure 16. Schematic diagram of the air duct of different models: (a) Original model; (b) Optimization model.

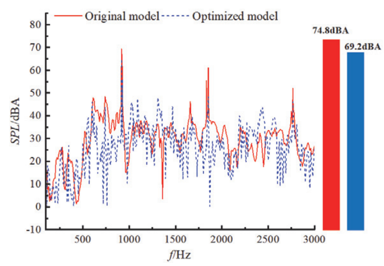
The acoustic finite element model of the optimal scheme was developed, which consistents with the original model. A comparative analysis of the SPL spectrum at monitoring points for both the optimization and original models was conducted, with a particular focus on the principal sound propagation paths. The SPL spectrum at monitoring point p_1, located on the air inlet side of the pile, and monitoring point p_2, situated on the air outlet side, were subjected to an in-depth comparison, as illustrated in Figures 17(a) and 17(b).

Figure 17 reveals that the main frequency of the optimization model corresponds to the BPF, with a notable decrease in the peak value of this frequency compared to the original model. This reduction is especially pronounced at the harmonic frequencies of the BPF, where the optimization model demonstrates a significant decrease. This is attributed to the shorter wavelength of high frequencies, which results in increased reflection and energy dissipation within the air ducts of the optimization model.At monitoring point p_1, the OASPL of the optimization model has been reduced from 79.8 dBA to 72.4 dBA, reflecting a substantial decrease of 7.4 dBA. Similarly, at monitoring point p_2, the OASPL has been lowered from 74.8 dBA in the original model to 69.2 dBA in the optimization model, indicating a reduction of 5.6 dBA.

(a)



(b)



(c)

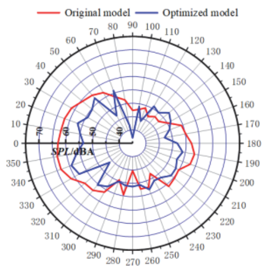


Figure 17. Results of acoustic monitoring points for different models: (a) Comparison of SPL spectrum and OASPL at point p1 obtained by simulation; (b) Comparison of SPL spectrum and OASPL at point p2 obtained by simulation; (c) Comparison of sound radiation directivity obtained by simulation.

Comparing the sound radiation directivity distributions of the two models at BPF and examining Figure 17(c), it is evident that the optimization model demonstrates a reduction in far-field sound radiation compared to the original model, particularly noticeable in the air inlet and outlet directions. Furthermore, the sound radiation directivity pattern of the optimization model lacks the pronounced directivity characteristics of the original model, displaying a more uniform radiation capacity across all directions.

When comparing the SPL distribution maps of the two models in the center section along the ventilation direction, as depicted in Figure 18(a), a wider range of high SPL regions is observed in the external sound field of the inlet and outlet wind sides of the original model due to its vent facing the primary sound source propagation direction. The radiation intensity and range are greater on the inlet side than the outlet side. Figure 18(b) illustrates that the optimization model effectively reduces the propagation of sound waves at BPF. Most of the energy resides in the upper compartment of the pile. As the sound wave transmits to the lower compartment of the pile, the SPL diminishes, leading to a smaller overall outward propagation of SPL compared to the original model. Additionally, due to the obstructive nature of the optimization model in the strongly directional propagation path of the sound source, the external sound radiation range and energy are significantly diminished.

(a) (b)

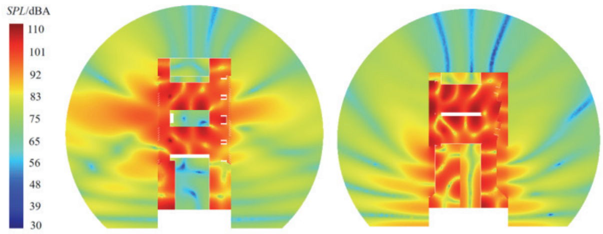
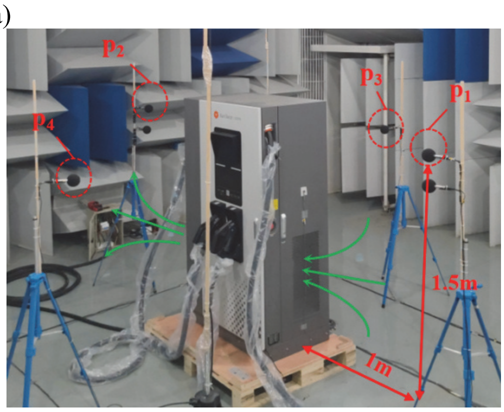


Figure 18. SPL distribution maps at BPF: (a) Original model; (b) Optimization model.

4.2. Experimental Validation of The Pile Acoustic Optimization Scheme

The optimization scheme was also tested and verified, and the microphone placement is exactly the same as the original model, as shown in Figure 19(a). Under the full power operating condition of the module, the spectral results of the optimization scheme at each monitoring point are shown in Figure 19(b), which is compared with Figure12(b) above and shows that the acoustic eigenfrequency distribution of the optimization model does not change compared to that of the original model, but the amplitude has been reduced more significantly. Compared to the other monitoring points, the amplitude of SPL at point p_1 is still the largest in the whole frequency band, but the difference between each monitoring point is reduced. Therefore, the optimization model is not as strong as the original model in terms of directivity of the sound radiation.

(a)



(b)

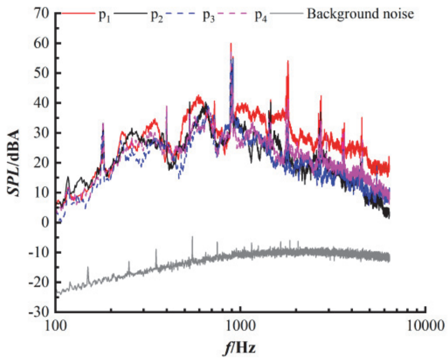


Figure 19. Acoustic monitoring points and spectrogram of the optimization model: (a) Experimental site and microphone layout of the optimization model; (b) Experimental Spectrum of SPL at monitoring microphones of the optimization model.

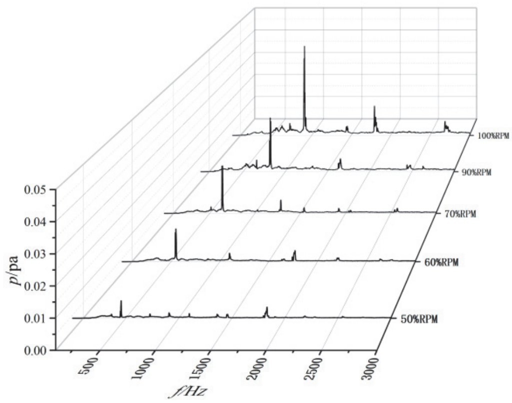
The comparison between the original and optimization models in terms of OASPL, as measured at each point, is summarized in Table 1 below. Notably, at monitoring point p_1, the OASPL is reduced by 6.8 dBA, from 77.6 dBA in the original model to 70.8 dBA in the optimization model. Similarly, at point p_2, the reduction is 3.9 dBA, from 69.4 dBA to 65.8 dBA. Conversely, at points not along the primary sound propagation paths, changes in OASPL are minimal. These findings underscore the effectiveness of the optimization model in reducing OASPL along the primary sound propagation path of the pile.

Table 2. Comparison of the measured OASPL at each monitoring point of different models.

Monitoring point	Original model result (dBA)	Optimization model result(dBA)	Difference value (dBA)
p_1	77.6	70.8	6.8
P_2	69.4	65.5	3.9
P_3	63.7	64.7	-1
P_4	64.3	65.2	-0.9

Comparing the sound pressure spectrum of the original model and the optimization model at the significant noise monitoring point p_1 under different rotational speeds of the module fan, as shown in Figure 20. The noise characteristic frequencies of the two models are all their BPFs and harmonic frequencies under the rotational speeds of the fan at that time, and there are no other frequency peaks highlighted, which indicates that the noise of the pile is caused by the aerodynamic noise of the fan. As the speed increases, the location of the BPF gradually moves to the right and the amplitude shows a linear trend of gradually increasing. It is worth noting that the amplitude of the optimization model is lower than that of the original model under all operating conditions, and the reduction in the peak at the BPF is more significant under high speed conditions. This is due to the fact that as the speed increases, the BPF gradually increases and the wavelength gradually decreases, resulting in more intense reflection loss at the BPF for the optimization model.

(a)



(b)

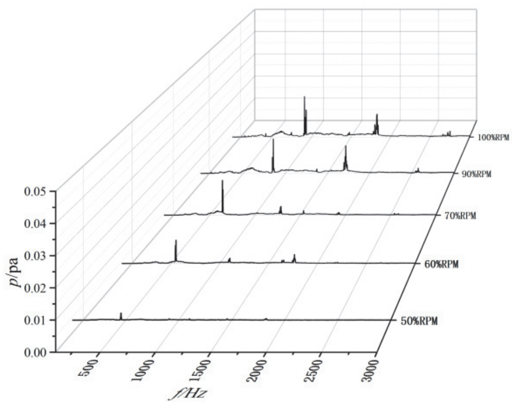


Figure 20. The measured sound pressure spectrum diagram under different conditions of different models at monitoring point p_1: (a) Original model; (b) Optimization model.

5. CONCLUSION

In this paper, a 120kW DC charging pile was taken as the research object, and CFD coupled CAA calculation method was used to analyze the sound source characteristics of the power supply module, which is the sound source component of the whole pile, and its propagation characteristics, and at the same time, combined with the experimental test of a single module, to validate the accuracy of the simulation, and then the sound propagation simulation for the whole pile was carried out on the basis of this simulation, and at the same time, the simulation results of the whole pile noise were tested and verified again. Finally, by combining the frequency spectrum and propagation direction characteristics of the DC charging pile noise, the design was optimized for the air duct structure, which effectively reduces the radiated noise of the charging pile. The conclusions of the study are as follows:

(1) The simulation analysis of CFD coupled CAA for a single power module, the sound source of the module mainly contributed by the dipole source which comes from the wall surface of the blades of the internal cooling fan and the wall surface of the upstream and downstream components. Moreover it is the main source of the BPF and its harmonic frequencies amplitude, and it is propagated outward by the inlet and outlet, and it has a strong directivity. The simulation results are in good agreement with the measurement results, so the simulation method can accurately predict the generation of the module sound source and its propagation law.

(2) The simulation analysis of the sound propagation of the original charging pile model, the characteristic frequencies of the pile is obtained as the fan BPF and its harmonic frequencies, which is completely consistent with the spectrum law of the module, and it is still mainly the aerodynamic noise generated by the internal cooling fan. The noise is mainly propagated outward through the pile vent with strong directionality, and the original model straight ventilated air duct has less obstruction on the main sound propagation path, so it makes more noise radiated outward.

(3) A comparative analysis of the pile noise reduction scheme design reveals that the optimized model can effectively inhibit the noise propagation while ensuring the heat dissipation, especially for the BPF and its harmonic frequencies peaks have a more obvious reduction, and in the main noise radiation monitoring point of the pile, the OASPL of the optimization model is reduced from 79.8dBA of the original model to 72.4dBA, which is a reduction of 7.4dBA.

(4) The analysis of the pile experiment comparing the original model and the optimization model reveals that the sound spectrum of the optimization model mirrors that of the original model, with a consistent distribution of characteristic frequencies aligning closely with simulation results. Under different speed conditions, the predominant frequency for both models is the fan BPF, wherein the peak value of the optimization model at the BPF decreases across all conditions, most noticeably under maximum speed conditions. The OASPL at main sound radiation monitoring points for both models demonstrates strong alignment with experimental measurements, with the absolute value error of the OASPL at these points not exceeding 3%. Furthermore, the disparity between the noise reduction values predicted by the simulation for the optimization model and the actual measurements is minimal, showcasing a mere 0.6 dBA deviation.

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